

Biodiversity

LESSON 6

Objectives

At the end of the lesson, students should be able to:

1. **Explain** the different levels of biodiversity.
2. **Discuss** the ecological, economic, social, and scientific importance of biodiversity.
3. **Identify and analyze** major threats to biodiversity.
4. **Describe and evaluate** strategies for conserving biodiversity.

What is Biodiversity?

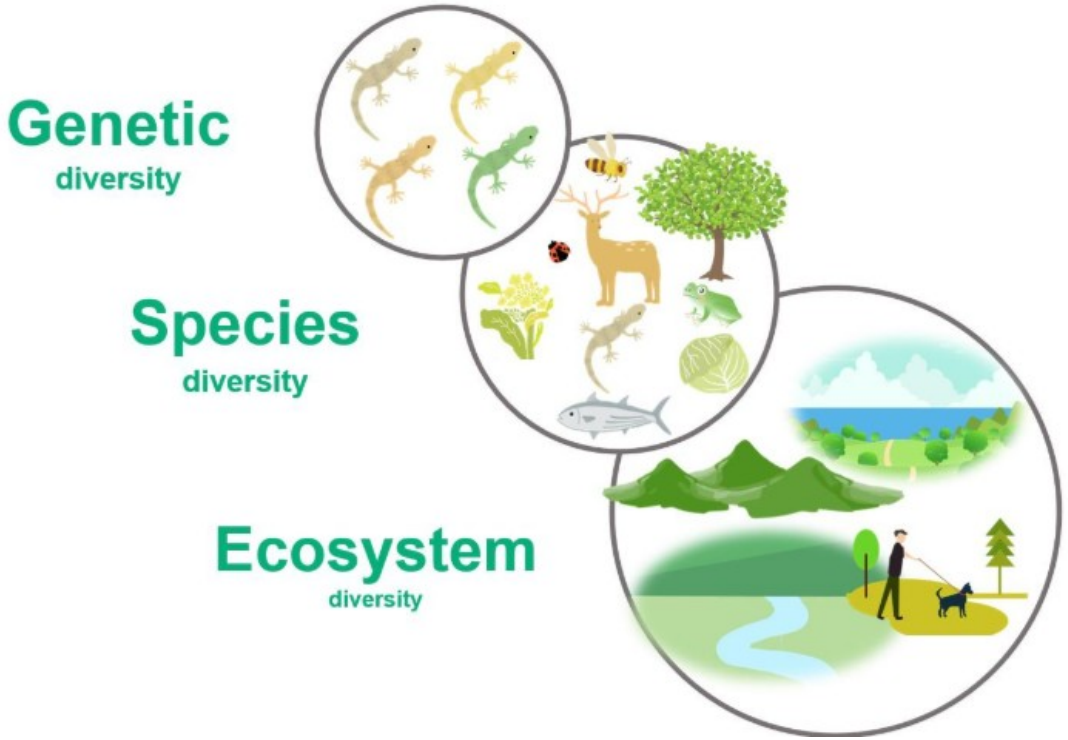
The term **biodiversity** comes from the words **biological** and **diversity**.

Biodiversity is the variety and variability of living organisms and the ecosystems in which they occur.

Levels of Biodiversity

Biodiversity can be studied at **three major levels**:

1. **Genetic diversity**
2. **Species diversity**
3. **Ecosystem diversity**



Genetic Diversity

- **Genetic diversity** refers to the differences in genetic characteristics among individuals of the same species.
- Individuals belonging to the same species are not necessarily genetically identical.

Example

- Rice varieties can differ in:
  - Grain size
  - Color
  - Taste
  - Resistance to pests
  - Tolerance to drought
  - Growth period

Why is genetic diversity important?

- Genetic diversity allows populations to adapt to:
  - Diseases
  - Climate change
  - Environmental changes
  - Pests
  - Other environmental pressures
- A population with greater genetic diversity generally has a greater range of characteristics that may help some individuals survive changing conditions.

Species Diversity

Species diversity refers to the variety of species present in a particular area.

- It involves two related ideas:

1. Species richness

- The **number of different species** in an area.

Example:

- Forest A contains 50 different species, while Forest B contains 20 species.
- Forest A has greater **species richness**.

2. Species evenness

- The **relative abundance** of different species in an ecosystem.
- Example:
- Two forests may contain the same number of species, but one may be dominated by only one species while the other has relatively balanced populations.

Example

- A coral reef may contain:
  - Hundreds of fish species
  - Corals
  - Sea stars
  - Sea urchins
  - Mollusks
  - Crustaceans
  - Algae
  - Microorganisms
- This represents high **species diversity**.

Ecosystem Diversity

Ecosystem diversity refers to the variety of ecosystems, habitats, and ecological communities within a geographic area.

Examples:

- Tropical rainforests
- Mangrove forests
- Coral reefs
- Seagrass beds
- Lakes
- Wetlands
- Grasslands
- Agricultural ecosystems

Level	Meaning	Example
Genetic diversity	**Variation** within a species	Different varieties of rice.
Species diversity	**Variety** of species in an area	Different species in a forest.
Ecosystem diversity	**Variety** of ecosystems in a region	Forests, mangroves, coral reefs.

Importance of Biodiversity

- Biodiversity is essential for both **ecosystem functioning and human well-being**.

A. Ecological Importance

B. Economic Importance

C. Food Security

D. Medicinal and Scientific Importance

E. Ecosystem Services

F. Cultural and Social Importance

A. Ecological Importance

- Biodiversity helps **ecosystems function properly**.
- Different organisms perform different ecological roles.

Example:

- Plants produce oxygen and organic matter.
- Pollinators help plants reproduce.
- Decomposers recycle nutrients.
- Predators regulate populations.
- Soil organisms help maintain soil fertility.

B. Economic Importance

- Biodiversity provides resources that support economic activities.

Examples:

  - Agriculture
  - Fisheries
  - Forestry
  - Ecotourism
  - Pharmaceuticals
  - Food production

- Many products used by humans originate from biological resources.

Example

- Fish populations support:
  - Fisherfolk
  - Fish vendors
  - Food processors
  - Restaurants
  - Consumers
- Biodiversity can contribute directly to **livelihoods and economic development**.

C. Food Security

- Biodiversity contributes to food security by providing a variety of:
  - Crops
  - Livestock
  - Fish
  - Fruits
  - Vegetables
  - Wild food resources
- Genetic diversity in crops is particularly important because it can provide varieties that are resistant to diseases, pests, drought, or other environmental stresses.

D. Medicinal and Scientific Importance

- Many organisms contain compounds that can be used in medicine and scientific research.
- Plants, fungi, bacteria, and marine organisms have contributed to the discovery of biologically active substances.
- Biodiversity therefore provides a large **natural resource for scientific discovery**.

E. Ecosystem Services

Ecosystem Service	Example
Provisioning	Food, freshwater, timber, medicines
Regulating	Climate regulation, flood control, pollination.
Cultural	Recreation, tourism, spiritual and cultural values.
Supporting	Nutrient cycling, soil formation, primary production.

F. Cultural and Social Importance

- Biodiversity is closely connected to human culture.
- Plants, animals, landscapes, forests, rivers, and coastal ecosystems may have:
  - Cultural significance (Philippine eagle)
  - Traditional uses (Bamboo)
  - Recreational value (Coral reefs)
  - Educational value (Mangrove forests)
  - Aesthetic value (The Banaue Rice Terraces)
- For many communities, natural resources are also connected to traditional knowledge and livelihoods.

Threats to Biodiversity

- Biodiversity is declining because of both **human activities and natural environmental changes**.

- **Major threats**
  - **H** – Habitat loss
  - **I** – Invasive species
  - **P** – Pollution
  - **P** – Population/human pressure
  - **O** – Overexploitation
  - **C** – Climate change

Habitat Loss and Destruction

- **Habitat loss** occurs when natural habitats are destroyed or converted into areas used for human activities.
- Examples:
  - Deforestation
  - Urbanization
  - Mining
  - Agricultural expansion
  - Road construction
  - Coastal development

Example

- When a forest is cleared for agriculture, organisms that depend on the forest may lose:
  - Food
  - Shelter
  - Breeding sites
  - Protection
- Some species may be forced to move, while others may decline or disappear.

2. Invasive Species

- An **invasive species** is a non-native organism that establishes itself and causes ecological, economic, or other harm.
- Such species may:
  - Compete with native species
  - Consume native organisms
  - Introduce diseases
  - Alter habitats
  - Disrupt food webs

Example

- The introduction of non-native species into an ecosystem can change relationships among native organisms and reduce native biodiversity.

3. Pollution

Pollution can damage organisms and their habitats.

- **Air pollution**
  - Can affect plants, animals, and atmospheric processes.
- **Water pollution** can result from
  - Sewage
  - Agricultural runoff
  - Industrial waste
  - Plastics
  - Oil

Cont.

- **Soil pollution** can result from:
  - Pesticides
  - Heavy metals
  - Improper waste disposal
- **Plastic pollution**
  - Plastic waste can enter rivers and oceans and harm marine organisms through ingestion or entanglement.

4. Overexploitation

- **Overexploitation** occurs when organisms or natural resources are harvested faster than they can naturally reproduce or recover.

Examples:

- Overfishing
- Illegal logging
- Excessive hunting
- Collection of wild plants
- Wildlife trafficking

Example

- If fish are caught faster than they reproduce, the fish population can decline.

6. Climate Change

- Climate change can alter environmental conditions such as:
  - Temperature
  - Rainfall
  - Sea level
  - Ocean temperature
  - Ocean chemistry

- Species may need to:
  - Adapt
  - Move to new habitats
  - Change their behavior
- Species that cannot adapt or move quickly enough may experience population decline.

Example

- Coral reefs are particularly vulnerable to increasing ocean temperatures because prolonged heat stress can cause **coral bleaching**.

Conservation of Biodiversity

Biodiversity conservation refers to the protection, management, and sustainable use of biological diversity.

- Major approaches:
  - In-situ conservation
  - Ex-situ conservation
  - Sustainable use
  - Habitat restoration
  - Laws and policies
  - Community participation
  - Education and awareness

A. In-Situ Conservation

- **In-situ conservation** means conserving organisms **in their natural habitats**.

Examples:

- National parks
- Wildlife sanctuaries
- Marine protected areas
- Forest reserves
- Protected landscapes and seascapes

Example

- Protecting a coral reef through a **marine protected area** allows organisms to remain within their natural ecosystem.

B. Ex-Situ Conservation

- **Ex-situ conservation** means conserving organisms **outside their natural habitats**.

Examples:

- Zoos
- Botanical gardens
- Seed banks
- Gene banks
- Aquariums
- Captive breeding facilities

Example

- A rare plant may be cultivated in a botanical garden when its natural population is threatened.

C. Sustainable Use of Biological Resources

- **Sustainable use** means using biological resources at a rate that allows them to remain available for future generations.

Examples:

- Sustainable fishing
- Sustainable forestry
- Responsible wildlife management
- Sustainable agriculture
- Ecotourism

Example

Instead of catching unlimited fish, fishing regulations can establish:

- Catch limits
- Fishing seasons
- Minimum size limits
- Protected areas
- Restrictions on destructive fishing methods

D. Habitat Restoration

- **Habitat restoration** involves repairing or improving ecosystems that have been degraded or damaged.

Examples:

- Reforestation
- Mangrove restoration
- Coral reef rehabilitation
- Wetland restoration
- River rehabilitation

Example

- Planting mangroves in degraded coastal areas can help restore habitat for fish, crabs, birds, and other organisms while also helping protect coastlines.

E. Laws and Policies

Governments establish laws and policies to protect biodiversity.

Republic Act No. 9147

- **Wildlife Resources Conservation and Protection Act**
- It provides for the conservation and protection of wildlife resources and their habitats.

Republic Act No. 7586

- **National Integrated Protected Areas System (NIPAS) Act of 1992**
- It established a system for identifying and managing protected areas.
- It was strengthened by:

Republic Act No. 11038

- **Expanded NIPAS Act of 2018**
- It expanded the country's protected-area system and strengthened protection and management provisions.

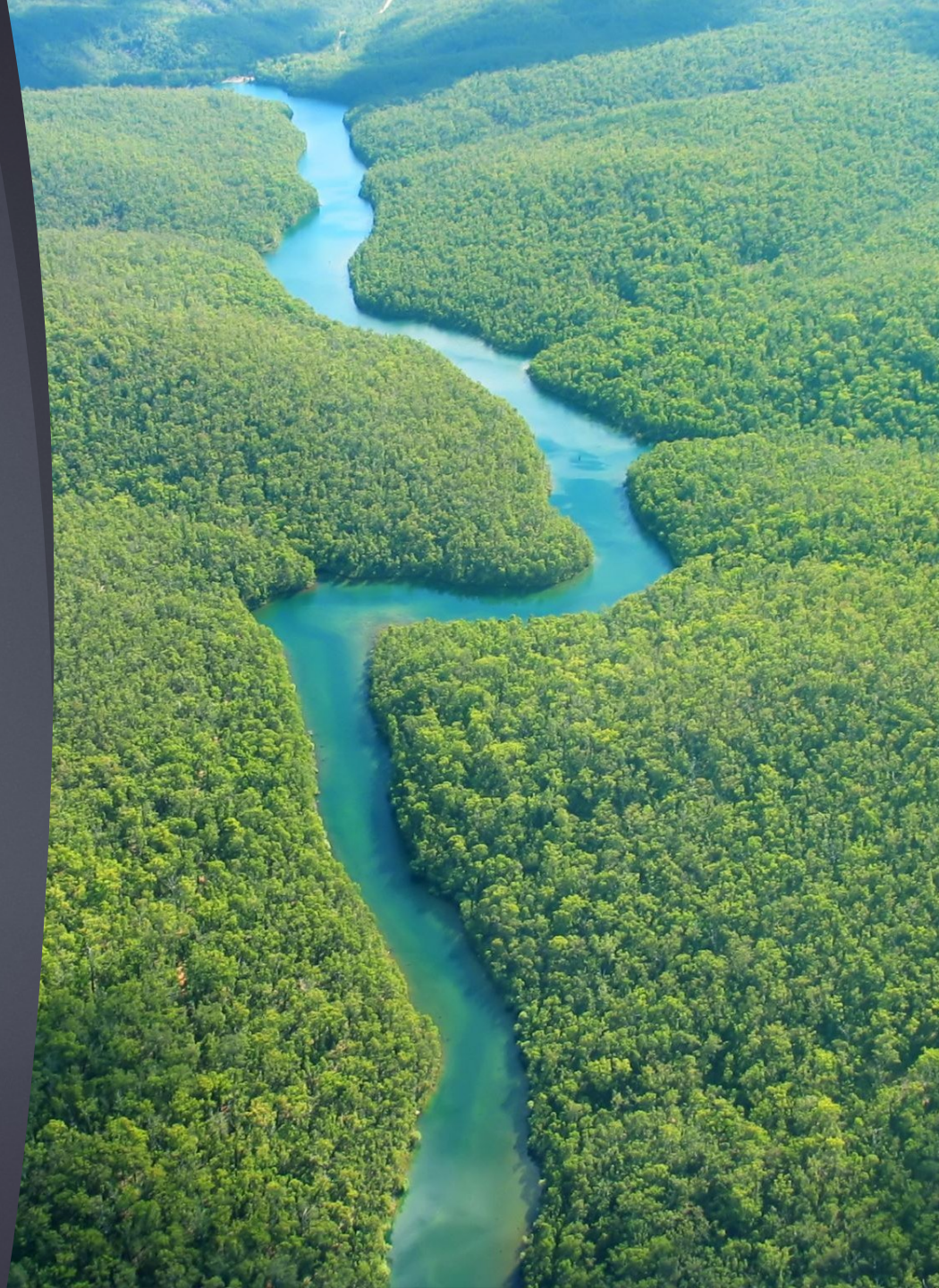
Biodiversity in the Philippines

- The Philippines is recognized as a **megadiverse country** because of its high biological diversity and high level of endemism.
- A **megadiverse country** is a country that has **an exceptionally large variety of living organisms (biodiversity)** and contains a **high number of species that are found nowhere else in the world (endemic species)**.
- Example: the **Philippine eagle** is endemic to the Philippines.
- Endemic species are particularly vulnerable because if their limited habitat is destroyed, they may have nowhere else to naturally occur.

Class Discussion

1. What would happen to human life if biodiversity continued to decline?

2. A community plans to convert a mangrove forest into a commercial development.
    a. What biodiversity-related consequences might occur?
    b. What alternatives could allow the community to achieve economic development while protecting biodiversity?



Key Concepts to Remember

Biodiversity	Variety of life and ecological systems
Genetic diversity	Variation within a species
Species diversity	Variety of species in an area
Ecosystem diversity	Variety of ecosystems
Species richness	Number of species in an area
Species evenness	Relative abundance of species
Endemic species	Naturally occurs only in a particular area
Habitat loss	Destruction or conversion of natural habitat
Habitat fragmentation	Division of continuous habitat into smaller patches
Overexploitation	Excessive use or harvesting of organisms
Invasive species	Non-native species that causes harm
Pollution	Introduction of harmful substances or energy
Climate change	Long-term changes in climate conditions
In-situ conservation	Conservation in the natural habitat
Ex-situ conservation	Conservation outside the natural habitat
Sustainable use	Using resources without preventing future availability
Habitat restoration	Repairing degraded ecosystems

Topic	Summary
Levels of Biodiversity	Biodiversity occurs at genetic, species, and ecosystem levels.
Importance	Biodiversity supports ecosystem functions, food production, livelihoods, medicine, culture, and ecosystem services.
Threats	Habitat loss, invasive species, pollution, overexploitation, and climate change threaten biodiversity.
Conservation	Biodiversity can be protected through in-situ and ex-situ conservation, sustainable use, habitat restoration, laws, community participation, and education.